

A PBL report on

Financial Management Tool

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C E R T I F I C A T E

This is to certify **that Aadya Lomte (B25IT1165), Shubham Pohakar (B25IT1180), Raunak Meshram (B25IT1182), Rohit Pravin Kadam (B25IT1184)**, have successfully completed the PBL Project entitled - “ **Financial Management Tool (FinPilot)**” under my supervision, in the partial fulfillment of First Year Engineering of Savitribai Phule Pune University.

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NOMENCLATURE

Symbol / Abbreviation		Meaning
UI	User Interface	
UX	User Experience	
PBL	Project Based Learning	
API	Application Programming Interface	
OCR	Optical Character Recognition	
UPI	Unified Payments Interface	

ABSTRACT

Managing personal finances, including budgeting, tracking spending, and saving for goals, can be challenging for many people. This project report describes a web-based financial management tool named FinPilot, designed to help users manage money more easily. The tool is prototyped using Figma, with a simple interface focused on everyday user needs.

Key features include pages for entering income and expenses, setting budgets and savings goals, and viewing clear charts and summaries of financial data. The dashboard shows total balance, recent transactions, and progress toward goals, while analytical screens display spending breakdowns and trends. The user interface uses a dark theme with green accents to ensure readability and visual comfort.

Even though this is a design prototype, it outlines how a real web app could work using technologies. The user interface prioritizes clarity and minimalism so users can quickly understand their finances. We used Figma components and links between frames to create an interactive prototype covering the complete user journey from login to analytics.

In summary, the project delivers a complete UI design for a Personal Finance Web App that addresses budgeting, expense tracking, and goal-setting in an intuitive way.

Keywords: Personal Finance, Budgeting, Expense Tracking, Figma, UI/UX Design, Prototyping, Financial Goals, Analytics

Chapter 1: Introduction

1.1 Background of the Study

Personal finance management involves planning budgets, tracking income and expenses, and saving for financial goals. It plays a key role in achieving financial stability. Studies show that effective budgeting and saving are essential pillars of personal financial stability. By setting a clear budget, individuals can monitor cash flow and reduce unnecessary spending, and by saving regularly they build reserves for emergencies or goals.

However, many people still find it difficult to manage money without a clear system. Without an easy way to record and review finances, people may overspend or fail to reach their savings goals. Common problems include losing receipts, forgetting to track small expenses, or lacking motivation to save. A recent consumer survey found that about 83 percent of adults use some type of digital finance app to help manage money. This shows strong interest in financial tools, but also suggests opportunities to improve them.

Digital tools are useful because they provide automatic calculations, visual summaries, and reminders. Unlike manual methods such as notebooks or spreadsheets, digital apps can update balances instantly and show charts that make spending patterns easy to understand. This project explores how a well-designed web application can make personal finance management simpler and more effective for everyday users.

1.2 About the Project

This project introduces FinPilot, a web-based personal finance management tool designed with a clear and simple interface. The name FinPilot reflects the idea of guiding users through their money management journey. Using Figma, we created a complete set of screens for budgeting, expense tracking, goal setting, and analytics, all aimed at everyday users.

The interface is designed to be minimal and shows only key information at a glance. Users can log in securely, view their financial summary on a dashboard, add income and expenses with categories, set monthly budgets, create savings goals, and view detailed analytics with charts and breakdowns. The design uses a dark theme with green accent colors to ensure visual comfort and readability.

The project was completed as a group of four students for the EEL2 course under the Department of Engineering Science and Humanities at Marathwada Mitra Mandal's College of Engineering. The work involved brainstorming, user research, wireframing, high-fidelity UI design, prototyping, and usability testing.

1.3 Objectives

The main objectives of this project are:

- To provide an easy way for users to record and track their income and expenses.
- To help users create and manage budgets for different spending categories such as food, bills, and entertainment.
- To enable users to set savings goals for emergencies, travel, or other purposes and monitor their progress.
- To present financial data including balances, spending summaries, and trends on a clear dashboard using charts and summaries.
- To design a clean, intuitive user interface using Figma that makes the tool accessible to non-technical users.
- To build a clickable Figma prototype showing the complete user flow from login to reports.

1.4 Scope of the Project

The scope of this project is focused on designing and prototyping the financial management tool interface. In scope are all the planned features: income and expense entry pages, budget setup, goal tracking, summary analytics, user profile management, notification settings, and payment method management. We include a login and signup system for multiple users, and settings pages for preferences.

The report covers system analysis, UI design, user research, and prototype flow. The scope excludes backend implementation and live data integration. We do not build a real database or connect to banks. Also outside the scope are advanced algorithms or machine learning for

financial advice. Instead, this project provides a full design blueprint in Figma that could later guide development of a working web app.

Chapter 2: Understanding User Needs

2.1 Survey Design & Methodology

To understand how people manage their finances, our team designed a short informal survey. The survey included both multiple-choice and open-ended questions about daily spending habits, budgeting methods, and challenges. We distributed the survey to classmates, friends, and family members through online messaging platforms.

The survey was kept short to encourage participation. It asked about the frequency of expense tracking, tools currently used, biggest challenges in money management, and interest in a simple finance app. A total of 25 responses were collected, which provided enough insight to guide our design decisions.

2.2 Data Collection

The collected data showed that most respondents tracked expenses only occasionally or not at all. About 60 percent used manual methods like notebooks or mental notes, while 30 percent used mobile banking apps without dedicated budgeting features. Only 10 percent used specialized finance apps.

Common reasons for not tracking expenses included lack of time, finding it boring, and not knowing which tool to use. Many respondents expressed interest in a simple app that could show their spending at a glance without requiring too much setup. This reinforced our decision to prioritize simplicity and quick onboarding in the FinPilot design.

2.3 Data Analysis

From the survey responses, we identified three main user groups: students with limited income who need basic tracking, working professionals who want budget control, and homemakers who manage household expenses. Each group had slightly different needs, but all wanted clarity and ease of use.

We summarized the findings in a table to guide feature prioritization. The most requested features were expense tracking, monthly budget summaries, and savings goal tracking. Advanced features like investment tracking and bank integration were considered lower priority for the first version.

Feature	User Demand	Priority
Expense Tracking	High	High
Budget Summary	High	High
Goal Setting	Medium	High
Analytics Charts	Medium	Medium
Bill Reminders	Low	Medium
Bank Integration	Low	Low

Table 2.1: Survey Questionnaire Summary

2.4 Functional Requirements

Based on user needs analysis, we defined the following functional requirements for FinPilot:

- User Authentication: Login, signup, forgot password, and reset password functionality.
- Dashboard: Display total balance, income, expenses, recent transactions, and savings goal progress.
- Income Management: Add income with amount, category, date, and description fields.
- Expense Management: Add expenses with amount, category, date, description, and payment method.
- Expense History: View, filter, and search past transactions by date or category.
- Budgeting: Set monthly budgets for different categories and track spending against limits.
- Goal Setting: Create savings goals with target amounts, deadlines, and progress

tracking.

- Analytics: View cash flow analysis, category breakdown, and monthly comparison charts.
- Payment Methods: Manage linked bank accounts and cards for transaction tracking.
- Notifications: Configure alerts for bill reminders, goal milestones, and budget limits.
- Profile & Settings: Manage personal information, security settings, currency, language, and dark mode.

2.5 Non-Functional Requirements

The following non-functional requirements were considered to ensure quality:

- Usability: The interface must be simple enough for non-technical users to understand without training.
- Responsiveness: The design should adapt to different screen sizes, especially mobile phones and tablets.
- Performance: Screens should load quickly with minimal delay, even on slower internet connections.
- Security: User passwords and financial data must be protected with encryption and secure authentication.
- Accessibility: Text should be readable, color contrast should meet standards, and navigation should be intuitive.
- Consistency: All buttons, forms, and cards should follow a uniform style across every screen.

Chapter 3: Stakeholder Interviews

3.1 Interview Preparation

To gain deeper insight into real-world money management problems, our team conducted informal interviews with five individuals from different backgrounds. The participants included a college student, a software employee, a small business owner, a homemaker, and a retired person.

We prepared a set of open-ended questions focusing on daily financial routines, tools used, frustrations with current methods, and desired features in a finance app. Each interview lasted about 15 to 20 minutes and was conducted either in person or over a video call.

3.2 Interview Findings

The interviews revealed several common patterns. Most participants struggled with remembering small daily expenses, which added up to significant amounts by month-end. Several mentioned that existing apps were too complex, requiring too many steps to enter a simple transaction.

The college student wanted a quick way to track pocket money and hostel expenses. The working professional needed monthly budget summaries to plan investments. The homemaker requested category-wise spending breakdowns for groceries and utilities. The business owner wanted payment method tracking to separate personal and business expenses.

The retired person appreciated simple visuals like charts instead of numbers.

Stakeholder	Main Need	Suggested Feature
College Student	Track daily expenses	Quick add expense
Working Professional	Budget control	Monthly budget view
Small Business Owner	Separate business/personal	Payment method tags
Homemaker	Category summaries	Pie chart breakdown
Retired Person	Simple visuals	Large text, clear charts

Table 3.1: Stakeholder Interview Summary

3.3 Key Insights

From the interviews and survey, we extracted the following key insights that shaped our design:

- Speed matters: Users want to log expenses in under 10 seconds. A complicated form will discourage regular use.
- Visual summaries help: Charts and progress bars motivate users better than plain tables of numbers.
- Categories are essential: Grouping expenses into categories like food, transport, and bills helps users understand spending patterns.
- Goals drive behavior: Seeing a savings goal with a progress bar encourages users to save more.
- Security is expected: Users trust apps more when they see password protection and secure data handling mentioned.
- Notifications help: Gentle reminders about budgets and bills prevent overspending and missed payments.

Chapter 4: Design Concept & Ideation

4.1 Brainstorming Ideas

The team began by brainstorming the core idea after selecting topic P5, Financial Management Tool, from the given project list. We discussed common money management problems faced by people around us and thought of solutions a simple tool could offer. Each team member suggested features based on personal experience and the survey findings.

Initial ideas included expense tracking, budget alerts, savings goals, investment tracking, bill splitting, and receipt scanning. We listed all ideas on a shared document and then voted on which ones were most essential for a first version. We agreed to focus on the core features that would help the largest number of users: tracking income and expenses, setting budgets, and monitoring goals.

We also looked at existing finance apps for inspiration, noting what worked well and what felt overwhelming. This competitive analysis helped us decide to keep the interface minimal and only include features that directly support daily use.

4.2 Feature Planning

After brainstorming, we planned the specific features and their workflow. We made a detailed list of all necessary screens: welcome and login, signup, forgot password, dashboard, add income, add expense, expense history, savings goals, create goal, analytics, cash flow analysis, category breakdown, monthly comparison, profile, payment methods, settings, notification settings, and transaction details.

For each feature, we sketched rough ideas on paper. For example, we decided the dashboard would show total balance at the top, followed by quick action buttons for adding income and expenses, a recent transactions list, and a summary of savings goals. We also planned how categories like food, transport, and entertainment would appear in dropdown menus.

This planning stage ensured we covered all use cases and helped organize the later design work. We created a feature priority matrix to decide which screens to design first and which could be added later.

4.3 Final Concept Selection

Finally, we settled on the final concept and design direction. We named the tool FinPilot, short for Financial Pilot, to reflect guiding users through money management. We chose a dark theme with green and white accent colors because it looks modern, reduces eye strain, and makes financial numbers stand out clearly.

Our final concept combines the planned features into a cohesive app flow: a user opens the app, sees a welcome screen, logs in or signs up, reaches the dashboard, and can navigate to any feature through a bottom tab bar. The design language uses rounded cards, clear icons, and ample spacing to create a friendly and professional appearance.

We decided the app would target both mobile and web users, with the Figma prototype optimized for mobile screen sizes first since most users prefer tracking expenses on their phones. This mobile-first approach influenced our decision to keep forms short and buttons large for easy thumb interaction.

Chapter 5: User Personas & Journey Mapping

5.1 User Personas

To keep the design user-centered, we created three representative user personas based on our survey and interview data. These personas helped us make design decisions by asking what each user would need at every step.

Persona 1: Rahul, College Student

Rahul is 20 years old and lives in a college hostel. He receives a monthly allowance from his parents and does occasional freelance work. He struggles to track where his money goes by the end of each month. He needs a simple app that lets him log daily expenses quickly and see how much money remains for the month.

Persona 2: Priya, Working Professional

Priya is 28 years old and works in an IT company. She earns a regular salary and wants to save for a vacation and an emergency fund. She uses multiple payment methods including credit cards and UPI. She needs an app that categorizes her spending, sets budget limits, and reminds her before she overspends. Persona 3: Sunita, Homemaker

Sunita is 45 years old and manages household expenses for her family. She shops for groceries, pays utility bills, and tracks children's education costs. She prefers seeing spending summaries by category so she can plan the monthly household budget. She values a clear, large-text interface that is easy to read.

Name	Age	Occupation	Primary Need
Rahul	20	Student	Daily expense log
Priya	28	IT Employee	Budget & savings
Sunita	45	Homemaker	Category tracking

Table 5.1: User Persona Details

5.2 User Journey Maps

We created journey maps for each persona to understand their emotional state and needs at each stage of using a finance app. The journey covers five stages: awareness, onboarding, regular use, goal achievement, and long-term habit formation.

In the awareness stage, users feel confused about their spending and look for a simple solution. During onboarding, they want quick signup and an immediate view of their finances. In regular use, they need fast transaction entry and clear summaries. When approaching a goal, they feel motivated by progress bars. Over time, consistent use builds financial confidence.

The journey map helped us identify moments where the design could reduce frustration and increase delight. For example, we added a quick-add button on the dashboard because users in the regular use stage want minimal steps to log expenses.

5.3 Pain Points & Opportunities

From the personas and journey maps, we identified the following pain points and corresponding design opportunities:

- Pain Point: Manual tracking is tedious and often forgotten. Opportunity: Provide one-tap quick add with smart defaults for date and category.
- Pain Point: Existing apps are cluttered with ads and unnecessary features. Opportunity: Keep the interface clean with only essential information visible.
- Pain Point: Users do not understand where most money goes. Opportunity: Show a pie chart breakdown of spending by category with percentage labels.
- Pain Point: Goal tracking is abstract and unmotivating. Opportunity: Use progress bars with color coding and milestone celebrations.
- Pain Point: Multiple payment methods create confusion. Opportunity: Allow users to tag each transaction with a payment source.

Chapter 6: Task Flows & Information Architecture

6.1 Task Flow Diagrams

The task flow defines how a user completes common actions in the app. We mapped out the primary flows: logging in, adding an expense, checking the dashboard, creating a budget, setting a savings goal, and viewing analytics. Each flow was drawn as a step-by-step diagram to ensure logical navigation.

The main flow begins at the welcome screen where users can sign in or sign up. After authentication, the user lands on the dashboard. From the dashboard, the bottom navigation bar provides access to expenses, goals, analytics, and profile. Any screen can be exited using the back arrow to return to the previous screen.

For adding an expense, the flow is: Dashboard -> Tap Add Expense -> Fill form -> Save -> Return to Dashboard with updated balance. This short loop ensures users spend minimal time per transaction.

Figure 6.1 shows the complete task flow diagram from the Figma prototype, illustrating all navigation paths between screens.

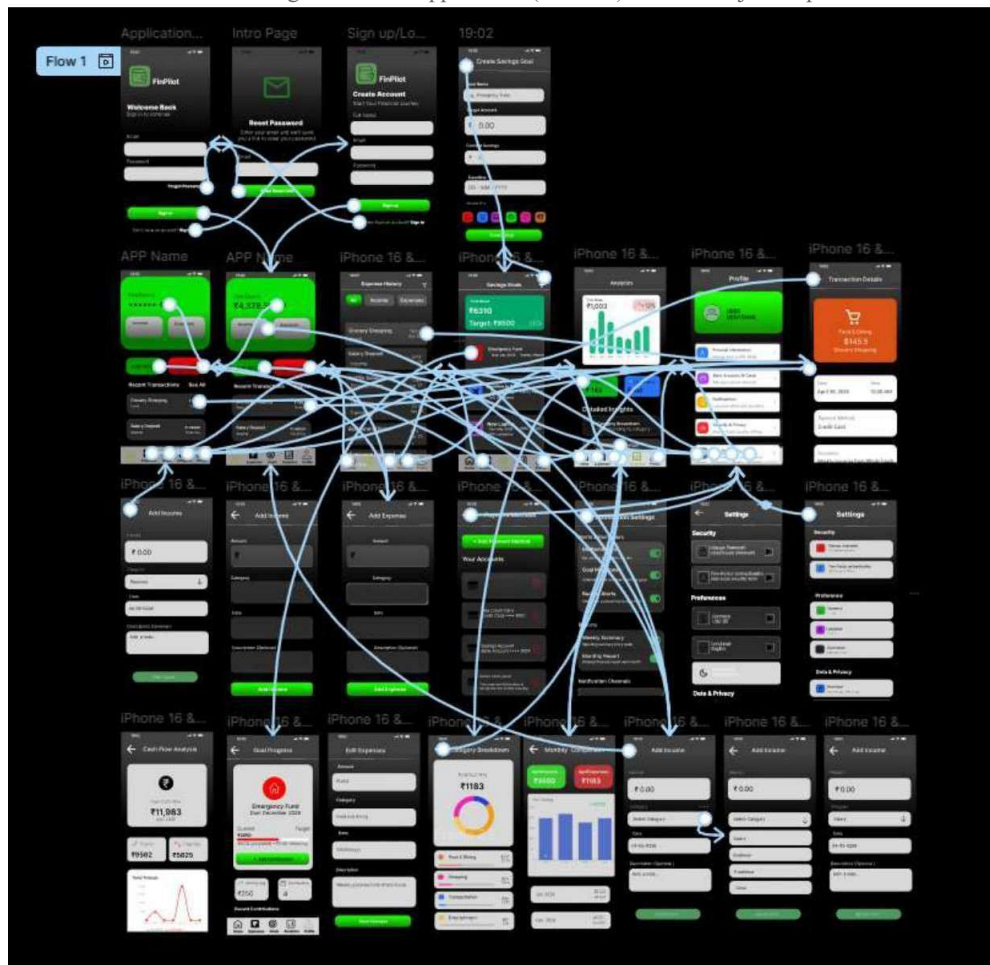


Figure 6.1: Complete Task Flow Diagram showing all screen connections in the FinPilot prototype

6.2 Sitemap / IA

The information architecture organizes all screens into a clear hierarchy. The app is divided into five main modules, each accessible from the bottom navigation bar on the dashboard.

- Home Module: Welcome, Login, Signup, Forgot Password, Reset Password, Dashboard.
- Expenses Module: Add Income, Add Expense, Expense History, Edit Expense, Transaction Details, Payment Methods.
- Goals Module: Savings Goals List, Create Savings Goal, Goal Progress, Add Contribution.
- Analytics Module: Analytics Overview, Cash Flow Analysis, Category Breakdown, Monthly Comparison.

- **Profile Module:** Profile Overview, Personal Information, Notification Settings, App Settings, Security & Privacy.

This modular structure keeps related features together and makes navigation predictable. Users always know which section they are in because the bottom tab highlights the active module.

Chapter 7: High-Fidelity UI Design

7.1 Design Approach

Our design approach focused on three principles: clarity, speed, and motivation. Clarity means every screen shows exactly what the user needs at that moment without clutter. Speed means common tasks like adding an expense require minimal taps and typing. Motivation means progress toward goals and budgets is visible and encouraging.

We chose a dark theme with green accents because it reduces eye strain during evening use, which is when many people review their daily spending. The green color symbolizes money and growth, while white text on dark backgrounds provides high contrast for readability.

All screens were designed in Figma using a consistent grid system and spacing scale. Buttons, input fields, and cards share the same corner radius and shadow style to create visual harmony. Icons were selected from a uniform icon library to ensure consistency.

7.2 UI Screens

This section presents the major screens of the FinPilot app with descriptions and figures.

Each figure shows a group of related screens from the Figma prototype.

7.2.1 Authentication and Goal Creation Screens

The welcome screen displays the FinPilot logo and app name, followed by the login screen where users enter email and password. A forgot password screen allows email-based reset. The signup screen collects full name, email, and password. These screens use simple forms with clear labels and green primary action buttons.

The create savings goal screen lets users define a goal name, target amount, current savings, deadline, and choose an icon. This structured form ensures users set realistic and trackable goals.

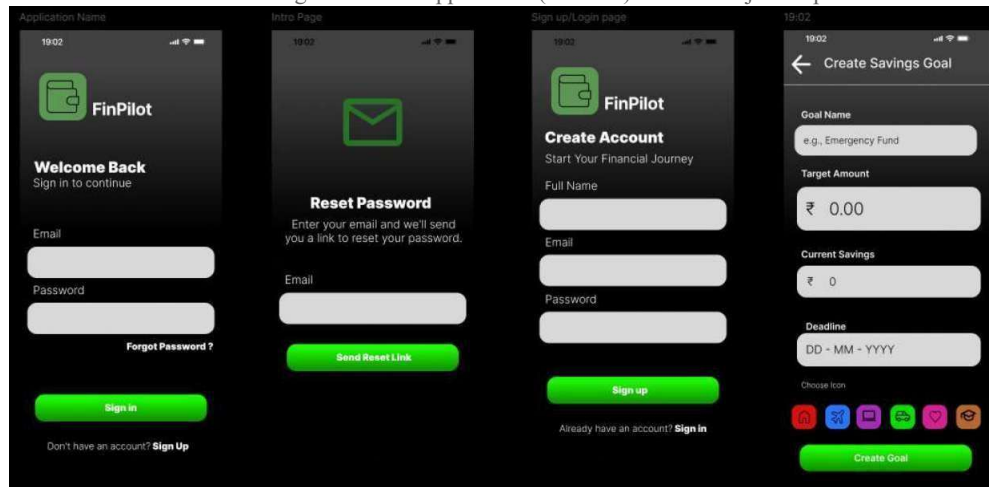


Figure 7.1: Login, Reset Password, Sign Up, and Create Savings Goal screens showing the authentication and goal setup flow

7.2.2 Dashboard and Transaction Screens

The dashboard is the home screen after login. It shows total balance at the top with an eye icon to hide or reveal the amount for privacy. Below are quick action buttons for income and expenses, followed by recent transactions and a bottom navigation bar.

The expense history screen lists all transactions with filters for all, income, or expenses. Each entry shows the title, category, date, and amount. The savings goals overview card displays total saved, target amount, and completion percentage.



Figure 7.2: Dashboard with balance display, Expense History with category filters, and Savings Goals overview card

7.2.3 Goals, Analytics, Profile and Details Screens

The savings goals list screen shows individual goals with icons, target amounts, deadlines, and completion percentages. The analytics screen presents a weekly bar chart, average daily spending, top category, and links to detailed insights.

The profile screen displays user information and provides access to personal details, payment methods, notifications, security settings, and app preferences. The transaction details screen shows full information for a single transaction including date, time, payment method, and

description.



Figure 7.3: Savings Goals list, Analytics dashboard, Profile menu, and Transaction Details screens

7.2.4 Income, Expense Entry and Payment Methods

The add income and add expense screens share a similar form layout with fields for amount, category, date, and optional description. A dropdown menu lets users select from predefined categories. The save button is prominently placed at the bottom in green.

The payment methods screen lists linked accounts and cards with masked numbers for security. Users can add new payment methods or delete existing ones. A security note reassures users that payment information is encrypted.

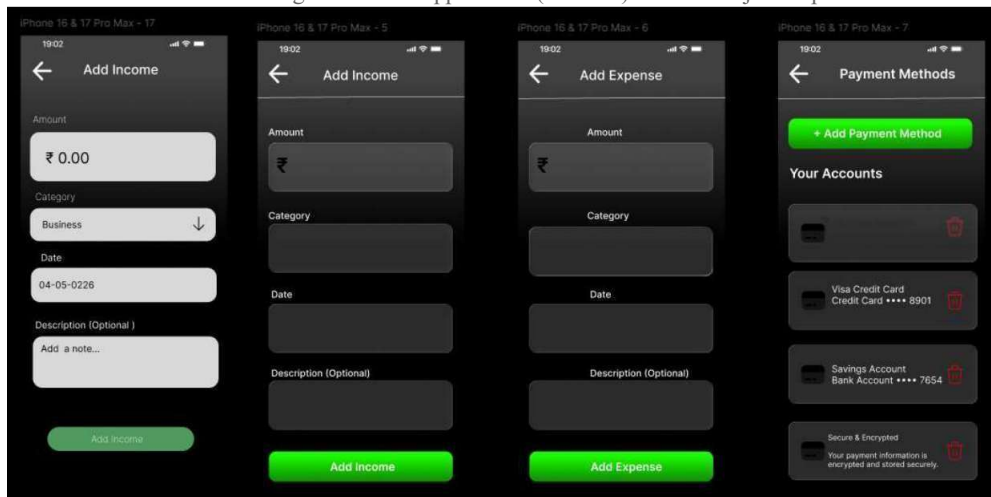


Figure 7.4: Add Income, Add Expense forms, and Payment Methods management screen

7.2.5 Notification and Application Settings

The notification settings screen uses toggle switches to control alerts for bill reminders, goal milestones, budget alerts, weekly summaries, and monthly reports. Users can also choose notification channels.

The settings screen is organized into sections: Security with password change and two-factor authentication, Preferences with currency, language, and dark mode options, and Data & Privacy with download and export options.

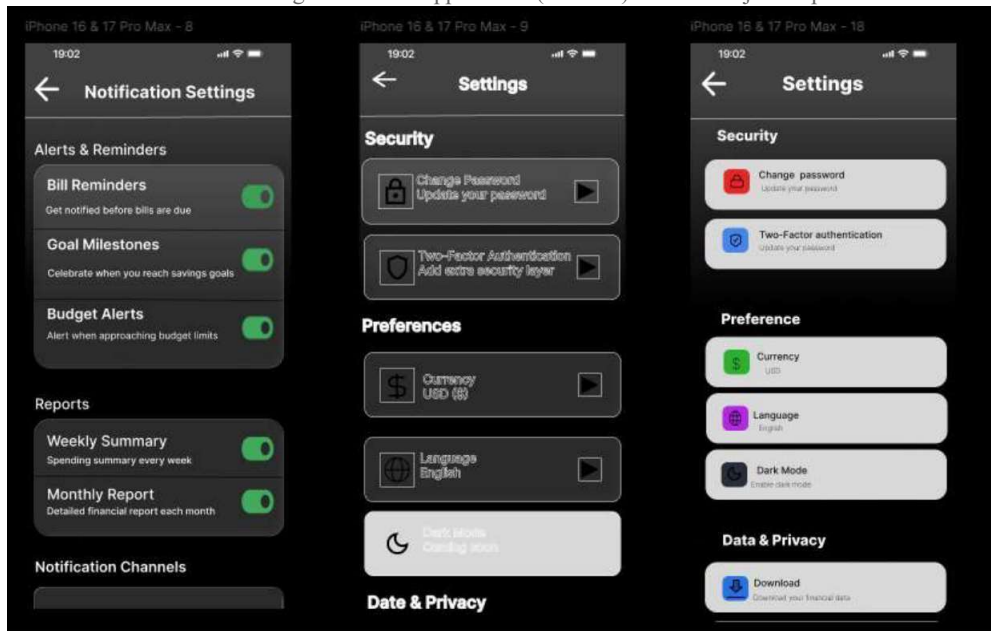


Figure 7.5: Notification Settings with toggle controls and App Settings with security, preferences, and privacy sections

7.2.6 Analytics and Goal Management Screens

The cash flow analysis screen shows net cash flow for a selected period with total inflow and outflow cards. A trend analysis chart plots income and expense trends over time.

The goal progress screen provides detailed view of a single goal with current amount, target, percentage completion, remaining amount, monthly average, and contribution history. Users can add contributions directly from this screen.

The edit expenses screen allows modification of existing transactions. The category breakdown screen displays a donut chart of spending by category with color-coded legends and percentages.

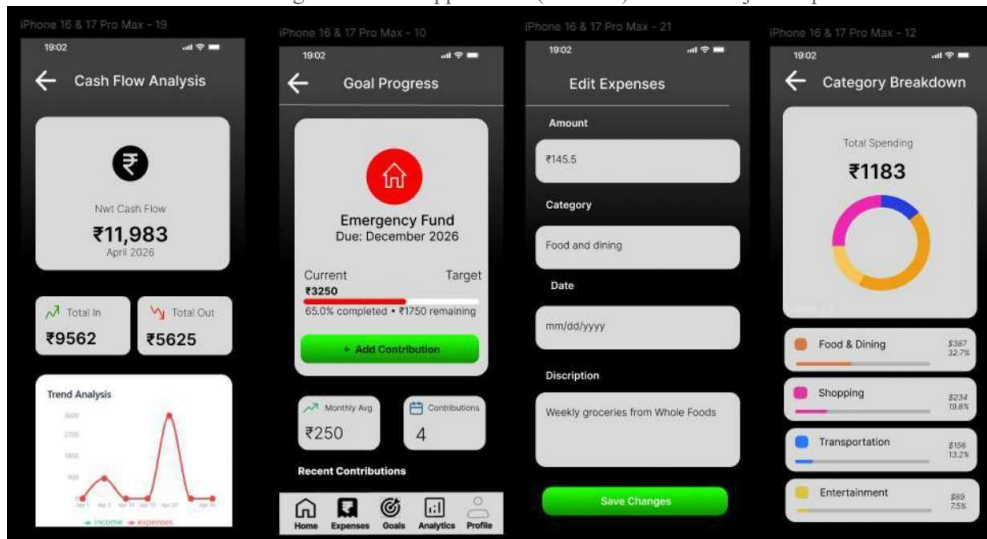


Figure 7.6: Cash Flow Analysis, Goal Progress, Edit Expenses, and Category Breakdown screens

7.2.7 Monthly Comparison and Category Selection

The monthly comparison screen compares income and expenses across months using a bar chart. It highlights net savings and shows month-wise saved amounts. This helps users identify spending trends and seasonal variations.

The add income screen includes a category dropdown that expands to show options like Salary, Business, Freelance, and Other. Selecting a category automatically fills the category field, reducing typing effort.

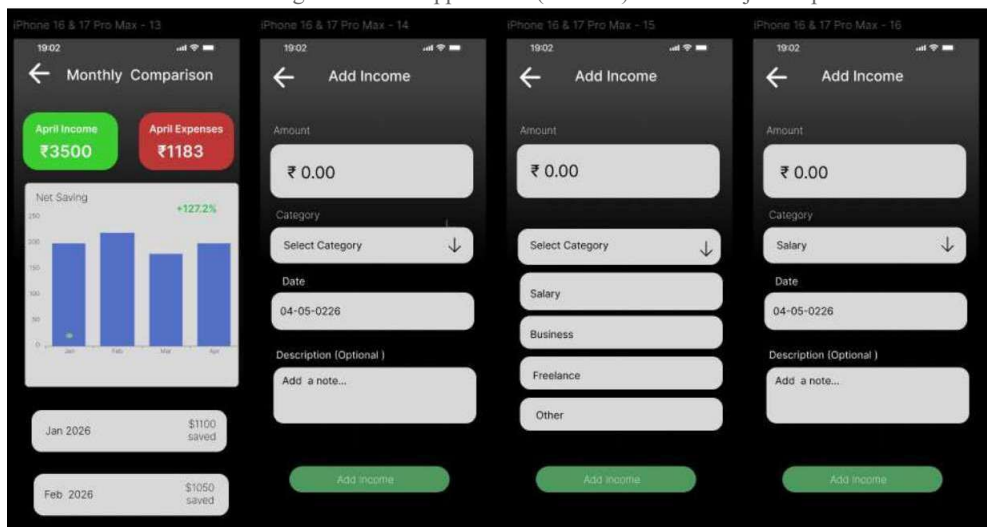


Figure 7.7: Monthly Comparison chart and Add Income screen with category dropdown selection

7.3 Interaction Design

Using Figma's prototyping features, we linked screens together to create an interactive prototype. Clicking the Add Expense button on the dashboard navigates to the expense form. Saving an expense returns the user to the updated dashboard. The bottom tab bar provides persistent navigation across all main sections.

Transitions between screens use simple slide animations to maintain a native app feel. Modal screens like transaction details and edit forms slide up from the bottom, while full screens like settings push in from the right.

Micro-interactions were added to enhance usability: buttons show a pressed state, toggle switches animate smoothly, and progress bars fill gradually when loading goal data. These small details make the prototype feel alive and responsive.

7.4 Iterations

The design went through multiple iterations based on team feedback. In the first version, the dashboard displayed too many numbers at once, which felt overwhelming. We reduced the initial view to show only total balance and recent transactions, moving detailed analytics to a separate screen.

The category selection was initially a text field, but feedback suggested a dropdown would be faster and more consistent. We changed to a dropdown with predefined categories. The color scheme also evolved: early versions used blue accents, but we switched to green to better match the financial theme.

Button placement was adjusted after testing. The original Add Income and Add Expense buttons were small icons, but we enlarged them to pill-shaped buttons with text labels for better accessibility.

Chapter 8: Style Guide Development

8.1 Typography

We selected a clean, readable font family for the app interface. For the Figma prototype, we used Inter as the primary typeface because it is designed for screens and offers excellent readability at small sizes. Headings use semi-bold weight at larger sizes, while body text uses regular weight.

The type scale is as follows: Screen titles at 24pt, section headings at 18pt, card titles at 16pt, body text at 14pt, and captions at 12pt. This hierarchy creates clear visual distinction between different levels of information.

For the project report itself, we use Times New Roman as required by the college format. All headings are bold, with chapter titles at 16pt, section headings at 14pt, and subsections at 12pt.

8.2 Color Scheme

The FinPilot color palette was chosen to convey trust, growth, and clarity. The primary colors are:

- Background: Dark charcoal (#1C1C1E) for the main app background, reducing eye strain.
- Surface: Slightly lighter dark gray (#2C2C2E) for cards and elevated surfaces.
- Primary Accent: Bright green (#34C759) for buttons, progress bars, and positive indicators.
- Secondary Accent: Teal (#30D5C8) for charts and alternative actions.
- Text Primary: White (#FFFFFF) for headings and main text on dark backgrounds.
- Text Secondary: Light gray (#8E8E93) for captions, placeholders, and disabled states.
- Expense Indicator: Red (#FF3B30) for expense amounts and overspending warnings.
- Income Indicator: Green (#34C759) for income amounts.

Color Name	Hex Code	Usage
Background	#1C1C1E	App background
Surface	#2C2C2E	Cards, sheets

Primary	#34C759	Buttons, progress
Secondary	#30D5C8	Charts, accents
Text Primary	#FFFFFF	Headings, labels
Text Secondary	#8E8E93	Captions, hints

Color Name	Hex Code	Usage
Expense Red	#FF3B30	Expense amounts

Table 8.1: Color Palette Specification

8.3 UI Components

We designed a library of reusable UI components to ensure consistency across all screens:

- **Buttons:** Primary buttons are green filled rectangles with rounded corners and white text. Secondary buttons are outlined. All buttons have a minimum height of 48pt for easy tapping.
- **Input Fields:** Text fields have a dark gray background, light border, rounded corners, and clear labels above. Dropdown fields show a chevron icon on the right.
- **Cards:** Information cards use the surface color with rounded corners and subtle padding. They group related information like transaction details or goal summaries.
- **Icons:** All icons come from a consistent icon set and use the text secondary color. Active navigation icons use the primary green color.
- **Progress Bars:** Goal progress uses horizontal bars with gradient fills from left to right. The percentage is displayed alongside the bar.
- **Charts:** Bar charts use the primary and secondary accent colors. Pie charts use a distinct color for each category to ensure clear differentiation.
- **Toggle Switches:** iOS-style switches with green active state and gray inactive state.

Chapter 9: Prototyping & Testing

9.1 Interactive Prototype

Using Figma's prototyping mode, we connected all screens to create a clickable prototype that simulates the full app experience. The prototype includes the following interaction flows:

- Authentication Flow: Welcome -> Login -> Dashboard, with branches to Signup and Forgot Password.
- Transaction Flow: Dashboard -> Add Income/Expense -> Form -> Save -> Dashboard with updated data.
- History Flow: Dashboard -> Expense History -> Transaction Details -> Edit or Delete -> History.
- Goals Flow: Dashboard -> Savings Goals -> Goal Progress -> Add Contribution -> Goals.
- Analytics Flow: Dashboard -> Analytics -> Category Breakdown / Monthly Comparison / Cash Flow.
- Settings Flow: Dashboard -> Profile -> Any Settings Screen -> Profile.

Each flow was tested internally by all four team members to ensure navigation felt natural and no dead ends existed.

9.2 User Testing

We conducted informal usability tests with five participants who had not seen the app before. Participants were given three tasks: log in and view the dashboard, add a new expense, and check the savings goal progress. We observed their actions and noted where they hesitated or made mistakes.

All five participants completed the login task without difficulty. Four out of five found the add expense form intuitive. One participant was initially unsure which button to press after filling the form, suggesting the save button needed stronger visual emphasis. Three participants praised the dashboard for being clean and easy to understand. Two participants suggested adding a search bar in the expense history for faster lookup.

Task	Success Rate	Key Feedback
------	--------------	--------------

Login	5/5	Straightforward
Add Expense	4/5	Save button needs emphasis
View Goals	5/5	Progress bar is motivating
Check Analytics	5/5	Charts are easy to read

Table 9.1: Usability Testing Results

9.3 Feedback & Improvements

Based on testing feedback, we made the following improvements to the design:

- **Save Button Emphasis:** Increased the size and brightness of the save button on all forms to make it more prominent.
- **Category Labels:** Added icons next to category names in the expense history for quicker visual scanning.
- **Dashboard Summary:** Added a small monthly summary card to the dashboard so users see their financial health without navigating to analytics.
- **Error States:** Added clear error messages for form validation, such as when amount is zero or category is missing.
- **Empty States:** Designed friendly empty state illustrations for screens with no data, encouraging users to add their first transaction or goal.

Chapter 10: Results & Discussion

10.1 Final Output

The final outcome of the project is a complete clickable prototype of the FinPilot financial management web app, designed in Figma. All major features are represented: authentication, transaction entry, expense history, budget monitoring, goal tracking, analytics, profile management, and settings.

The prototype contains over 25 unique screens connected by more than 50 interaction links. Users can log in, add income or expenses, and immediately see updated charts and balances on the dashboard. The design successfully combines all objectives: it provides a clear method to record transactions, follow budgets, and view goals.

Figure 10.1 shows the complete Figma prototype canvas with all screens and their connections.

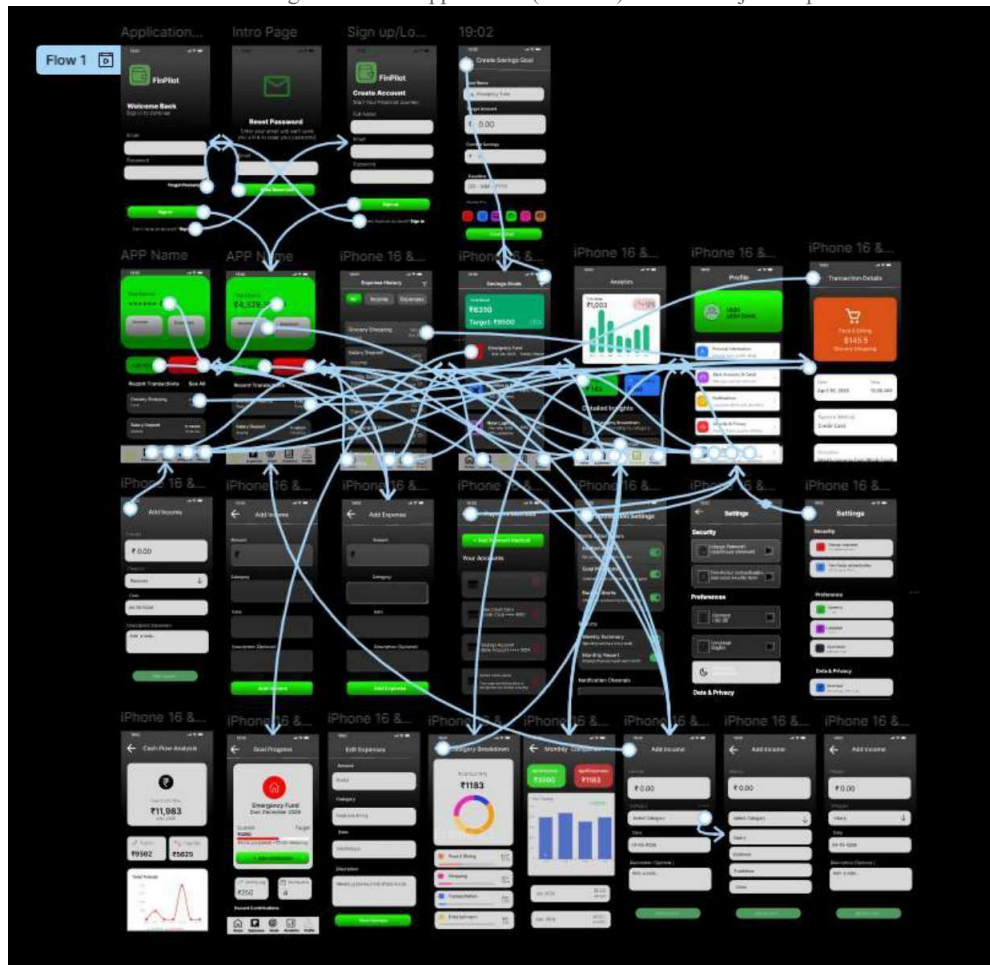


Figure 10.1: Final Figma prototype showing all screens and navigation connections

10.2 Design Impact

The FinPilot design addresses real user problems identified in our research. The simple transaction entry form reduces the barrier to consistent expense tracking. The dashboard summary gives users immediate awareness of their financial status. The goal progress visualization provides motivation to save.

Compared to existing solutions like spreadsheets and complex finance apps, FinPilot offers a middle ground: more structured than manual tracking but simpler than full-featured banking software. The dark theme and green accents create a distinctive visual identity that feels modern and trustworthy.

The project demonstrates that even a design prototype can validate user needs and guide future development. Feedback from testers confirmed that the core concept is sound and the interface is approachable for users with varying levels of financial literacy.

Chapter 11: Conclusion

11.1 Summary

This project successfully created a full-fledged design prototype of a personal financial management tool called FinPilot. The prototype covers all planned features including budgeting, expense tracking, goal setting, and financial analytics in a cohesive web interface. By focusing on simplicity and user needs, we ensured that the app presents financial data in a clear and organized way.

The team learned and applied important UI/UX principles such as keeping the interface uncluttered, using visual charts for easier understanding, and designing for mobile-first usage. We also gained practical experience with Figma, using its tools to build interactive screens, components, and prototyping links.

Overall, the project met its objectives by providing an intuitive design that helps everyday users take control of their finances. It lays the groundwork for future development by showing how the conceptual design can guide actual web app implementation.

11.2 Limitations

As a design-focused prototype, this project has certain limitations:

- The prototype has no real backend or database, so all data is simulated and not persisted between sessions.
- There is no connection to real bank accounts, UPI, or payment services for automatic transaction import.
- User authentication is visual only; no actual security mechanisms like encryption or token-based login are implemented.
- Advanced features such as AI-based spending insights, receipt scanning, and investment tracking are not included.
- The prototype is optimized for mobile screen sizes and may require additional design work for desktop or tablet layouts.

11.3 Future Scope

There are many ways to extend this project in the future:

- **Real Data Integration:** Connect the app to bank accounts or payment services via APIs so transactions can be imported automatically.
- **Mobile App Development:** Build native iOS and Android apps using frameworks like React Native or Flutter based on the Figma designs.
- **Notifications and Reminders:** Add automatic alerts for upcoming bills, budget overspending, and goal milestones through push notifications.
- **Advanced Analytics:** Use machine learning to predict future spending, suggest personalized budget adjustments, and detect unusual transactions.
- **Export and Sharing:** Allow users to export data as PDF or Excel reports, and share budgets with family members for collaborative planning.
- **Receipt Scanning:** Add OCR-based receipt scanning to automatically extract transaction details and reduce manual entry.
- **Multi-Currency Support:** Expand the app to support multiple currencies and exchange rate tracking for travelers.

Chapter 12: References

12.1 GitHub Repository Link

[Click here.](#) ([GitHub - aadyalomte2025it-spec/github](#) · [GitHub](#))

12.2 YouTube Link

[Click here.](#) (<https://youtu.be/wBkR4ovBbrw?si=QLvGQcLqk0OZPHre>)

12.3 References

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Chapter 13: Appendices

13.1 Survey Questionnaire

The following questions were used in our user needs survey:

1. How do you currently track your income and expenses? (Notebook / Spreadsheet / Mobile app / Mental notes / I don't track)
2. How often do you review your spending? (Daily / Weekly / Monthly / Rarely / Never)
3. What is your biggest challenge in managing money? (Forgetting expenses / Overspending / No clear budget / Lack of motivation / Other)
4. Which features would you want in a finance app? (Expense tracking / Budgeting / Goal setting / Analytics / Bill reminders)
5. Would you use a simple finance app if it helped you save money? (Yes / Maybe / No)

13.2 Interview Questions

The following questions were used during stakeholder interviews:

1. Walk me through how you managed your money last week.
2. What tools do you use to track income and expenses? What do you like or dislike about them?
3. Tell me about a time when you felt stressed about your finances. What caused it?
4. If you had a magic wand, what would you change about how you manage money?
5. What would make you trust a new finance app with your spending data?

13.3 Additional Design Screens

This appendix includes additional screenshots from the Figma prototype that support the main chapters.

Figure 13.1 shows the complete prototype flow diagram with all screen connections. Figures 13.2 through 13.8 show the full-resolution versions of the UI screens presented in Chapter 7.



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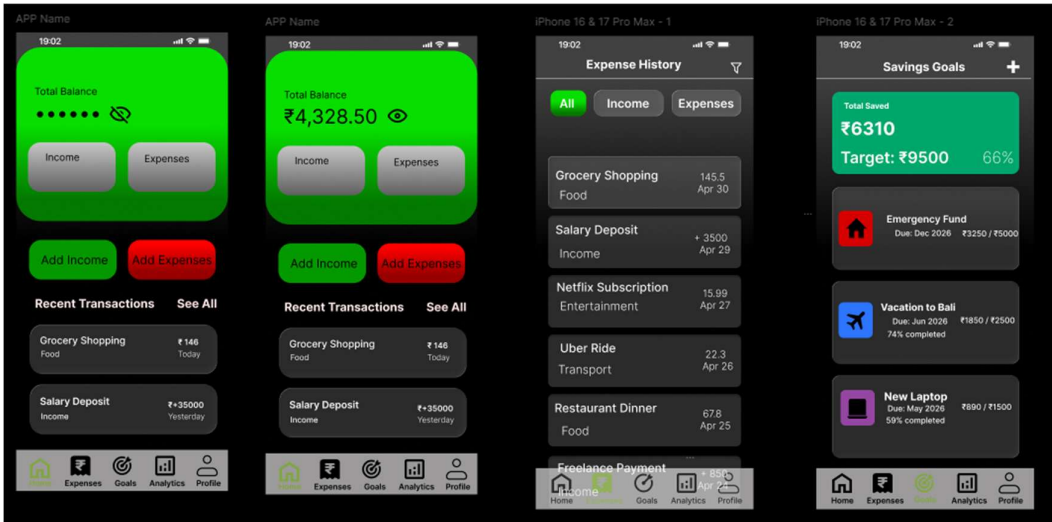
Website: www.mmcoe.edu.in

email id: mmcoe@mmcoe.edu.in

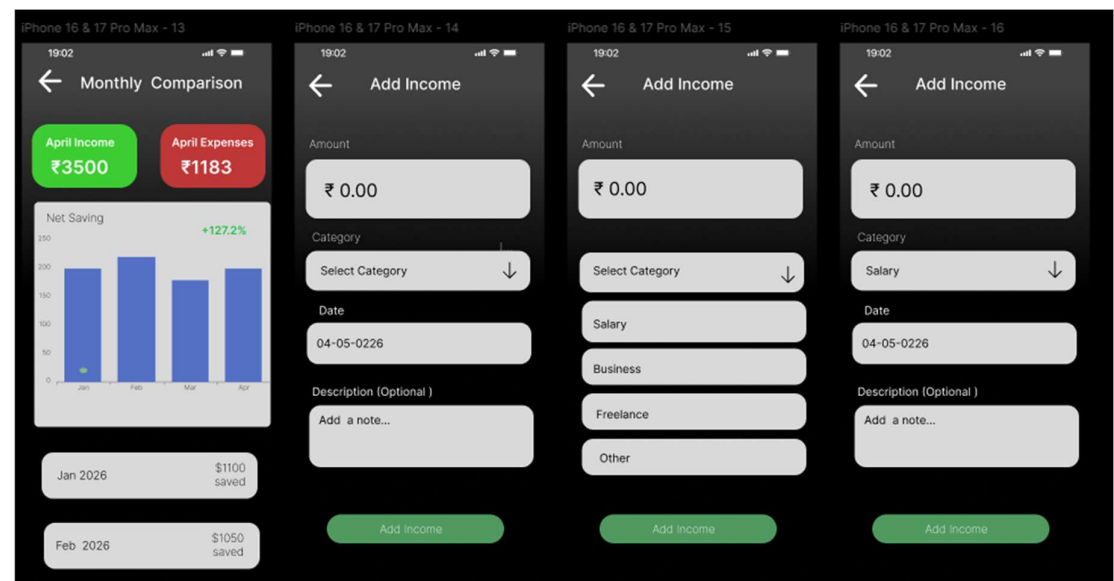
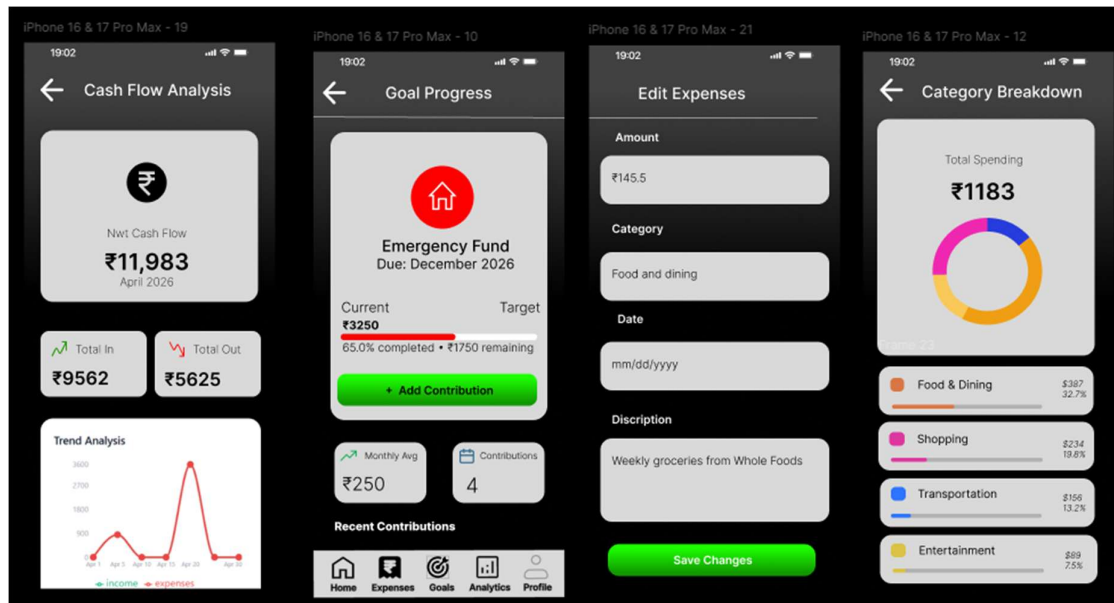
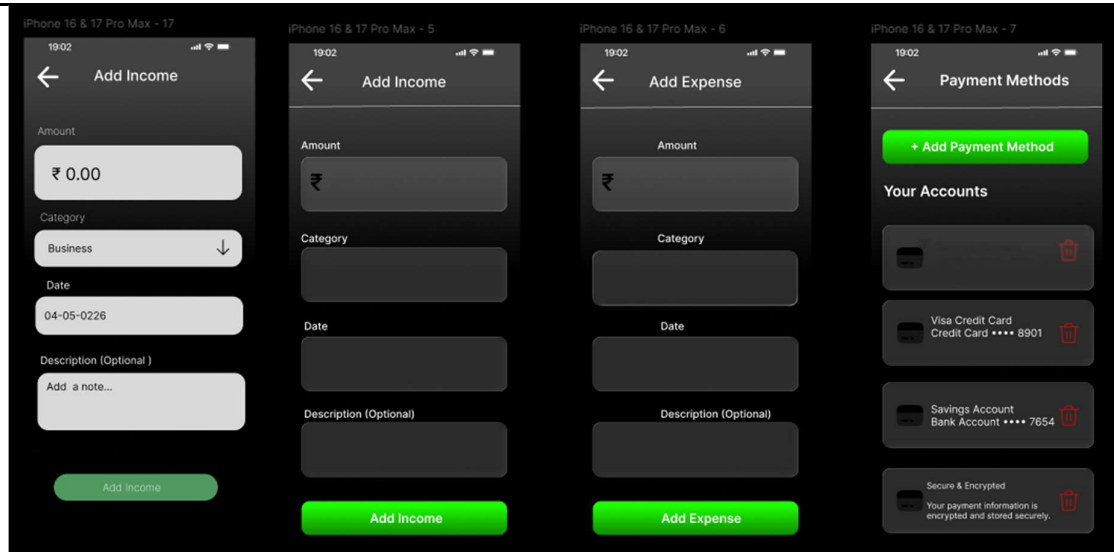
Department of Engineering Science and Humanities

Engineering Exploration Lab- II

P	
Project Title:	P5 - Financial Management Tool: [FinPilot] Develop a web-based tool for personal finance management, integrating budgeting, expense tracking, and financial goal setting with a clear and intuitive interface.
<u>Date</u>	<u>Assessment Points</u>
	Research : During the research phase, our team deployed online forms and questionnaires to collect real-world data from students and peers regarding their budgeting habits. The survey was carefully designed to identify the methods users currently rely on (such as mental math, physical diaries, or complex spreadsheets), the problems they encounter when trying to save money, and their expectations for a modern digital budgeting application. Key Survey Points Identified: • The need for a frictionless way to track daily micro-expenses and log monthly income without spending too much time. • A strong desire for a simple, visually appealing method to set, monitor, and achieve custom savings goals. • The expectation for clear visual analytics, such as charts and graphs, to make sense of monthly cash flows. • Essential technical requirements, including mobile-first responsiveness, strong data security, and automated push notifications for reminders.
	Analysis: Once the survey responses were collected, we conducted a thorough analysis to identify recurring themes and user frustrations. We categorized the feedback into three core areas: daily tracking, long-term budgeting, and goal setting. We realized that existing manual methods often lead to user burnout because they lack automation and visual feedback. Gaps Identified in Existing Solutions: • Cluttered Interfaces: Users suffer from a lack of a unified, intuitive dashboard that provides a quick snapshot of their financial health upon opening the app. • Tracking Friction: Manually categorizing and tracking small, daily expenses is tedious and often abandoned after a few days. • Goal Abandonment: There is a distinct absence of motivational tracking for long-term financial goals (like building an emergency fund or saving for a laptop), making it hard for users to stay disciplined.

	• Poor Data Visualization: Users struggle to understand their spending habits because they lack clear visualizations and category-wise breakdowns of their monthly cash flow.
	Ideate: Armed with the analyzed data, we brainstormed specific features and UI solutions to address these gaps. Our core philosophy for FinPilot was to make financial management visually engaging and effortless. We proposed creating a centralized dashboard for immediate balance checks, interactive progress bars for savings, and comprehensive yet easy-to-read analytics. Proposed Features for FinPilot: • A unified home screen displaying the total available balance, an immediate breakdown of recent transactions, and highly visible, quick-access buttons to "Add Income" or "Add Expense." • A dedicated "Savings Goals" module that allows users to create specific targets (e.g., Emergency Fund, Vacation to Bali) complete with target amounts, deadlines, and visual progress indicators. • A "Detailed Insights" and analytics section featuring cash flow charts and category breakdowns (e.g., Food & Dining, Shopping, Transport) so users can easily see where their money is going. • Advanced profile settings, including customizable UI preferences like Dark Mode, multi-currency support, and strict security measures like Two-Factor Authentication. 

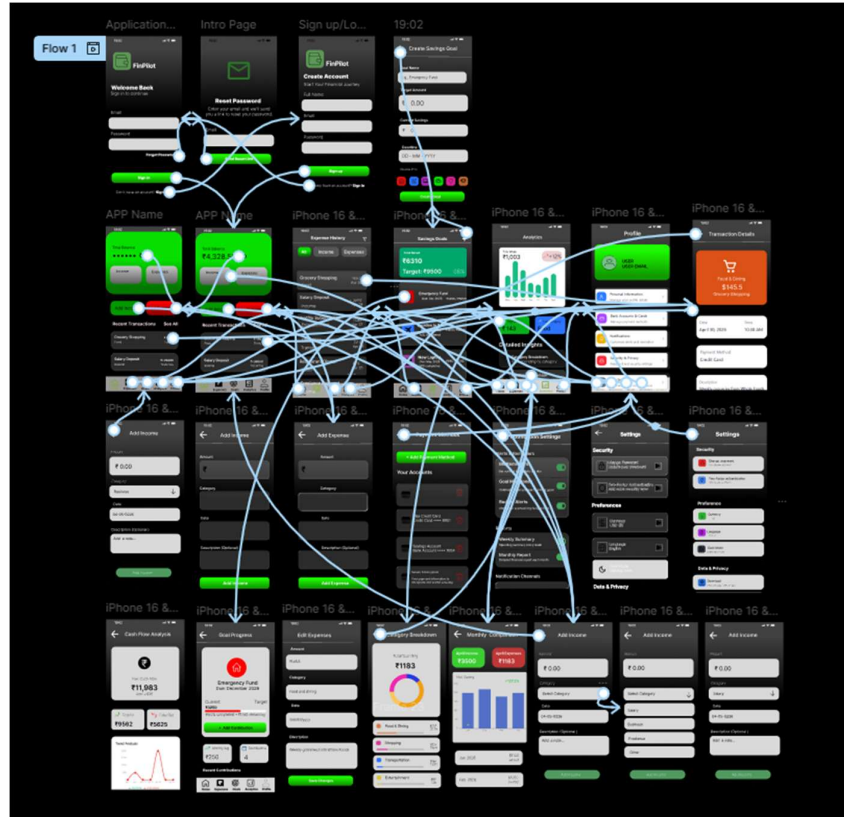
	Build: Moving into the build phase, we translated our functional and non-functional requirements into high-fidelity UI/UX wireframes using Figma. We meticulously designed the user flow to ensure that core actions—such as logging a transaction or checking a goal—could be done in minimal clicks. We built out dedicated screens for user onboarding, daily cash flow inputs, deep-dive analytics, and account management, ensuring the design remained cohesive, modern, and accessible.
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Test:

With the Figma prototype developed, we subjected our design to peer and faculty review. The goal was to verify the clarity of the interface, the completeness of the financial tracking features, and the overall relevance to the user pain points we identified earlier. We tested the navigational flow to ensure users

wouldn't get lost between checking their analytics and adding a new expense. Feedback from this phase was instrumental in refining button placements, color contrasts, and data readability before finalizing the requirements.



Implement:

We have successfully prepared the complete UI/UX design prototype via Figma and drafted the foundational Software Requirements Specification (SRS) document for the FinPilot platform. The structured design clearly defines how the system will look and operate to meet user needs.

Outcomes:

PRN	Name of Student	Remarks /Signature
B25IT1165 B25IT1180 B25IT1182 B25IT1184	Aadya Lomte Shubham Pohakar Raunak Meshram Rohit Pravin Kadam	
Summary :	This project encompassed the end-to-end design thinking process: from creating surveys and analyzing real-world data to gathering strict functional and non-functional requirements. Ultimately, we built empathy for the end-user, resulting in the creation of comprehensive, high-fidelity Figma prototypes for a robust personal finance management application aimed at improving financial literacy and habit-building	